

STILLSTACK · REQUIREMENTS

StillStack requirements

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DOCUMENT CONTROL

Document ID	SSK-REQ-001	Version	0.8
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stillstack		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Approved follow-ups carried into the design: R9 mass 18.7 kg, R12 USD 443.20 (USD 88.20 over the target), R4 lip modelled, R10 moves from at risk to met on paper (wick clips modelled)
0.7	2026-10-02	Amish Chadha	Decisions of 2026-10-02 noted under R4, R6, R7, R8 and R10 (lip, conductivity pass condition, fabric test, coating rule, PPS, wick clips); no status changed
0.6	2026-10-01	Amish Chadha	Status from the constructable design (SSK-DDR-003); R12 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Add TRL 3 status from SSK-CAL-001; record decisions from SSK-DDR-001 (budget kept at \$250, tilt stand, aperture); targets unchanged

0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

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These requirements were checked by calculation at TRL 3 in SSK-CAL-001 v0.6. No requirement is **not met**; five are at risk and one cannot be verified until TRL 4. The TRL 2 review decisions (SSK-DDR-001) kept the 1 m2 aperture and the 20 deg tilt on an adjustable stand. Under SSK-DDR-002 (decided by Amish on 2026-09-25, going with the recommendations) R8 was restated with a 150 °C rating for stages 1 and 2 plus a stagnation cover, and the R12 budget was raised from \$250 to \$320. The added materials and cover bring the parts to \$351.20. On 2026-09-26 Amish approved a budget top-up to \$355 (SSK-DDR-002 v0.2). On 2026-10-01 Amish set every budget as a value-engineering target rather than a limit, so R12 is reported against it. The constructable design of SSK-DDR-003 (made under Amish's 2026-09-30 instruction to make the design buildable; open for his review) was estimated at USD 416.20. With the approved follow-ups of 2026-10-02 (PPS rails, wick clips, labels) it is estimated at USD 443.20, USD 88.20 over the USD 355 target. Carrying the wick clips into the model moves R10 from at risk to met on paper; no other status changes.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (SSK-CAL-001)
R1	Produce more distillate per unit area than a basin still	10 L per m ² of aperture per day or more on a clear day with 5.5 kWh per m ² of sun (design goal 13 L)	Stage energy balance calculation; later outdoor test with a single-basin still alongside	Met on paper: 10.6 L after warm-up deduction (11.3 L quasi-steady); the 13 L goal is not met
R2	Reuse the latent heat of condensation	Daily gained output ratio (latent heat in distillate divided by solar input) of 1.3 or more	Stage energy balance calculation	At risk: 1.34 quasi-steady, 1.26 after warm-up deduction
R3	Remove salt	Distillate conductivity 75 µS/cm or less (about 50 mg/L dissolved solids) from feed up to 40 g/L salinity	Calculation of carryover paths; later conductivity meter	Not verifiable at TRL 3: brine carryover must stay below 0.075 % of the distillate (8.5 mL per day)
R4	Keep brine and distillate apart	A physical air break of 10 mm or more between every wick edge and every distillate channel; no shared drain	Design review of the model	At risk: 10 mm dry breaks at ribs, rails and inside the plate edges; brine tongues carry the wick tails 10.5 mm or more above the closed manifold lid and at least 44 mm from any distillate tongue (SSK-DDR-003 A1, accepted 2026-10-02); feed tails cross the condensing plates' high edges 6 mm above them, to be guarded by a 4 mm downturned lip on each condensing plate's high edge (A2 (c), decided 2026-10-02, now in the model as four tabs on each plate). Each stage's distillate conductivity test at TRL 4 is a pass condition before any water is drunk
R5	Run without power	No pumps, electronics or batteries; gravity feed from a user-filled container at most 1.5 m above ground	Design review	Met: fully passive; feed trough rim at most 1.00 m (at 35 deg)
R6	Accept real feed water	Seawater or brackish water up to 40 g/L salinity after cloth or 20 µm prefiltering; feed-to-distillate ratio 2.5 or more so salt stays in solution	Salt balance calculation	At risk: brine at 66.7 g/L is well below saturation, but the wick needs a permeability of 3.0×10^{-11} m ² or more to carry the feed, which is not yet known; the fabric is chosen by a wicking-rise test between a polyester-type nonwoven and a glass-fibre fabric (decided 2026-10-02)
R7	Food-safe wetted parts	Every surface that touches distillate is food-contact rated (for example 316 stainless steel, food-grade coated aluminum, PP, HDPE or NSF/ANSI 51 or 61 silicone)	Material list review	At risk: coated aluminum is decided, the coating class is selected (fluoropolymer, PTFE or PFA type, BOM line 5) but no named product with a drinking-water certificate for wet, salty service at 80 °C (and 150 °C dry on plates 1 and 2) has been seen. Rule decided 2026-10-02: the lowest-cost coating certified

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (SSK-CAL-001)
				for drinking-water contact and rated to 150 °C dry (fluoropolymer cookware coatings first), else 316 stainless for plates 1 and 2
R8	Survive dry stagnation	No damage after an empty-wick day at 1,000 W/m ² and 35 °C ambient, in calm air or wind: materials in stages 1 and 2 (absorber to plate 2, including wicks, rails, ribs, coating and liner) rated to 150 °C or more; all other internal materials rated to 110 °C or more; a stagnation cover supplied and a shade rule in the instructions (restated under SSK-DDR-002)	Stagnation temperature calculation; material datasheets	At risk: stages 1 and 2 reach 144 °C in calm air (6 K margin); stage 3 reaches 110 °C in calm air (0 K margin); cover specified (BOM line 14); rail polymer decided as PPS (2026-10-02, BOM line 6); coating class selected (fluoropolymer) but its drinking-water certificate not yet seen
R9	Household scale and portable	1.0 m ² aperture; panel mass 20 kg or less dry; carried by two people; footprint within 1.3 x 1.3 m	Massing model and mass estimate	Met: 18.7 kg dry (22.7 kg wet); footprint 1.28 x 1.25 m at 10 deg
R10	Easy maintenance	Wicks removable, rinsed and refitted by one person without special tools in 30 min or less; salt flushing no more often than once a week	Design review	Met on paper: the plates lift out once the low wall cap is off, and each wick is held by two stainless clips, at its high end and on its brine tongue, with no adhesive (decided 2026-10-02; in the model, SSK-DWG-119 and 120), so the wicks lift out; the 30 minute time is to be shown at TRL 4
R11	Adjustable tilt	Tilt between 10 and 35 deg to suit latitudes of about 0 to 35 deg	Model check	Met: props pinned at six holes, 10 to 35 deg in 5 deg steps; model built and checked at each
R12	Low cost and buildable	Parts cost against a value-engineering target of USD 355 for the 1 m ² prototype (budget_usd; a hypothetical control target, not a limit, Amish 2026-10-01); hand tools, sheet metal shears and a 3D printer only	Priced BOM	USD 88.20 over the value-engineering target: estimated USD 443.20 for the constructable design (SSK-DDR-003, with the PPS rails, wick clips and labels of 2026-10-02)

■ Assumptions

- Design day: 5.5 kWh per m² per day (19.8 MJ per m²) on the tilted aperture, 30 °C mean daytime ambient, light wind (1 m/s in SSK-CAL-001).
- Latent heat of vaporization: SSK-CAL-001 uses temperature-dependent values (about 2.33 to 2.36 MJ per kg over the stage range) in place of the flat 2.26 MJ per kg used at TRL 2.
- The WHO does not set a health-based limit for dissolved solids; 50 mg/L is chosen as a check that salt carryover is negligible, not as a potability limit.

- Household need of about 3 L per person per day for drinking and 2 L for cooking, so 10 to 11 L per day serves a household of two.